MATH 101 Unit 4: Exponentials, Logarithms, and Systems - Practice 3

Version 202610122020.45 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10.

(a) $\log(0.5) + \log(20000)$

(b) $\dfrac{\log_5(64)}{\log_5(16)}$

2. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain.

$f(x) = \log_4(x-3) + 3$

3. (a) Solve the linear system. (b) Solve the nonlinear system algebraically. (c) Test whether (0,0) belongs to the region given by the inequalities; show both comparisons.

(a) $x + y = 7$

$x - y = 1$

(b) $x^2 + y^2 = 17$

$y = x + 3$

(c) $x^2 + y^2 \le 17,\ y > x + 3$

4. Solve both equations exactly. Check the logarithmic domain in (b).

(a) $4^{x-3} = 2^{x+4}$

(b) $\log_4(x-3) + \log_4(5) = 2$

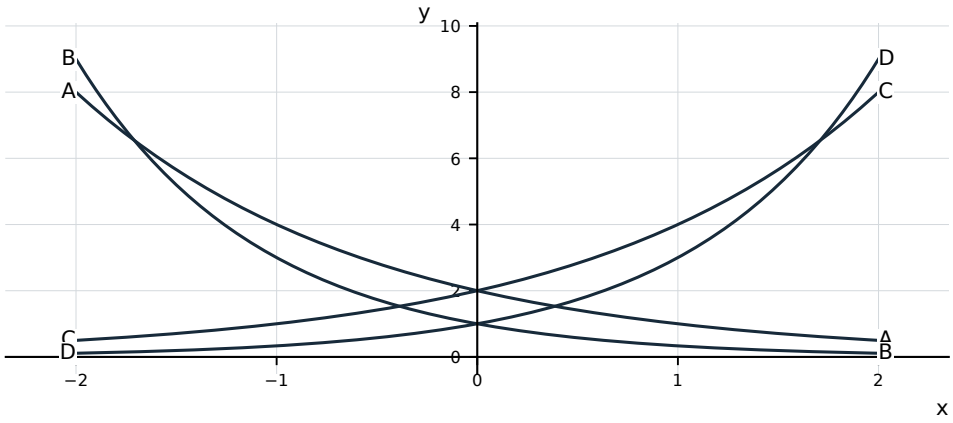
5. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 2(2)^x$

$(b)\quad 2(\frac{1}{2})^x$

$(c)\quad 1(3)^x$

$(d)\quad 1(\frac{1}{3})^x$



Student answer sheet

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Question 1

(a) 10 pts

(b) 10 pts

Question 2

(a) Domain 4 pts

(a) Range 4 pts

(b) Asymptote 3 pts

(b) x-intercept 3 pts

(c) Left end 3 pts

(c) Right end 3 pts

Student answer sheet

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Question 3

(a) Ordered pair

6 pts

(b) Intersections

10 pts

(c) Membership

4 pts

Question 4

(a)

8 pts

(b), with domain

12 pts

Question 5

(a)

5 pts

(b)

5 pts

(c)

5 pts

(d)

5 pts

Answer key

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Question 1

(a)	4	10 pts
(b)	$\frac{3}{2}$	10 pts

Question 2

(a) Domain	$(3, \infty)$	4 pts
(a) Range	$(-\infty, \infty)$	4 pts
(b) Asymptote	$x = 3$	3 pts
(b) x-intercept	$(\frac{193}{64},\ 0)$	3 pts
(c) Left end	$x \to 3^+ :\ f(x) \to -\infty$	3 pts
(c) Right end	$x \to \infty :\ f(x) \to \infty$	3 pts

Answer key

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Question 3

(a) Ordered pair	$(4,\ 3)$	6 pts
(b) Intersections	$(1,\ 4);\ (-4,\ -1)$	10 pts
(c) Membership	At (0,0): 0 <= 17, but 0 > 3 is false; not in the region.	4 pts

Question 4

(a)	$x = 10$	8 pts
(b), with domain	$x = \frac{31}{5};\ x > 3$	12 pts

Question 5

(a)	C	5 pts
(b)	A	5 pts
(c)	D	5 pts
(d)	B	5 pts